CS 330 Final Project: Design Decisions

Development Choices

For my final project, I created a 3D mug scene using C++, OpenGL, and the provided graphics framework. I used simple shapes to build the mug and added textures, lighting, and transformations to improve its appearance.

The mug body uses a cylinder, and the handle uses a torus. I placed the handle beside the body and adjusted its size and rotation to look like a real mug handle. I also added a plane underneath the mug to represent a wooden surface. Scaling, rotation, and translation helped me position each object correctly.

I applied stainless-steel texturing to the mug, gold to the handle, and wood to the surface. I also set up different material properties so each object would respond to lighting in its own way.

For lighting, I used multiple light sources and the Phong lighting model. Ambient lighting provides general illumination, diffuse lighting adds shading, and specular lighting creates highlights. Different light colors helped make the scene more detailed.

Navigation

I added camera controls to let users explore the scene from different angles. The WASD keys move the camera forward, backward, left, and right, while Q and E move it up and down.

Mouse movement changes the camera's viewing direction. Horizontal movement adjusts yaw, and vertical movement adjusts pitch. I also added mouse scrolling to change movement speed, making navigation easier to control.

The program supports both perspective and orthographic projection. Pressing P switches to perspective view, while O switches to orthographic view. Perspective gives the scene a stronger sense of depth, whereas orthographic projection displays objects without perspective distortion.

Custom Functions and Code Organization

I organized the code into separate classes and functions to make it easier to understand and update. In the SceneManager class, PrepareScene() loads the meshes and textures and prepares the materials and lighting. The RenderScene() function draws the objects using their assigned transformations, textures, and materials.

The DefineObjectMaterials() function sets the material properties for the scene, while SetupSceneLights() configures the lights, including their positions, colors, and intensity. Keeping these tasks separate makes it easier to adjust the scene's appearance without changing unrelated code.

The ShaderManager class handles loading and compiling the vertex and fragment shaders. The ViewManager class manages camera input and projection settings, while the Camera class controls movement and viewing direction. This organization reduces repeated code and makes individual features easier to maintain.

Conclusion

This project helped me understand how shapes, transformations, textures, lighting, and camera controls work together in a 3D environment. I also learned the importance of separating code into functions and classes. Testing different materials, lighting settings, and camera views helped me improve the scene and understand the graphics framework better.